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CO1-A13

1) Intelligent Agents in a smart Home Automation System:

An intelligent system that perceives its environment through sensors and takes action through actulators to achieve a desired objects

In a smart home, sensors may detect

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- Motion or human pressure.
- Temperature
- light level
- Time of day.
- Door/window status
- User activities and preferences.

The agent then controls devices such as lights, fans, air conditioners.

1. Simple Reflex Agent:

A simple reflex agent chooses an action based only on the current percept using condition -action rule.

Structure:

If condition → perform action

2. Model-Based Agent:

A model based agent maintains an internal model / state of the environment. It uses current sensor information together with information about previous situations.

3. Goal-Based Agent:

A goal based agent selects actions based on a desired goal.

For a smart home, goals could include:

1. Maintain comfortable
2. Reduce electricity consumption
3. Keep the house secure.
4. Turn off un-necessary appliances

4. Utility-Based Agent:

A utility-based agent chooses the action that provides the highest utility, meaning the best overall outcome according to several criteria:

1. Comfort
2. Energy efficiency
3. Security
4. Costs

29) Robot Solving an Unknown Maze Using UCS and IDS

Consider a robot placed in an unknown maze. The robot does not known the maze layout or the location of the exit beforehand. It must explore different paths until it finds a path to goal.

This is a search problem where:

1. Initial state → Robot's starting position
2. Actions → Move up, down, left, right.
3. States → Different positions in the maze.
4. Goal State → Exit of the maze.
5. Path cost → cost of movements.

Two uniformed search strategies can be used are

1. Uniform Cost Search.
2. Iterative Deepening search.

Uniform cost search:

Uniform cost search expands the node with the lowest total path cost.

It uses a priority queue.

If the robot has two possible paths.

- Path A = cost 5
- Path B = cost 10.

US UCS explores Path A first

Iterative Deepening search (IDS):

Iterative Deeping search repetedly performs depth-limited search, increasing the depth limit each time.

For example:

Depth 0 → Depth 1 → Depth 2 → Depth - 3

Example:

Suppose the exit is 3 moves away.

IDS

- Depth 0
- Depth 1
- Depth 2
- Depth 3 → Exit found.